Lecture 4 — Time

Keeper, for Casey Koons and the team

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Lecture 4 — Time

The question

What is time, in a geometry that has none built in — and why does it run one way?

The object of Lecture 1 has ten real dimensions and a positive-definite metric. Nothing in it moves. There is no axis labelled t, and if there were, the symmetry group would move it somewhere else. Yet the physics we read off the object has a time — the Schrödinger evolution of Lecture 3 needs a parameter, and measurement in Lecture 3 was a *commit*, something that happens and cannot un-happen. This lecture says where that parameter comes from and why it has a direction. It is, in our own accounting, the one place where the program has a complete derivation with nothing imported: the generator, the arrow, and the two faces of time are all readings of one operator the geometry hands us.

For the reader in a hurry

Our shape has a circle at its centre — one of the two pieces of the group that leaves a point fixed. Nothing turns that circle by itself. But the *records* on the shape's edge (Lecture 3) can only be made in one order, and "how far the record-making has run" turns out to be exactly "how far around that circle we have gone." That is time: not a stage the play is performed on, but a count of what has been committed. It runs one way because the operator that turns the circle has a lowest energy and no highest — you can always run it forward and never backward. And it has a smallest step, about a sixth of a billionth of a billionth of a second, which is the time light takes to cross the radius of the hydrogen atom's smallest orbit.

The generator is forced, and it is not the obvious one

Recall $K = \mathrm{SO}(5) \times \mathrm{SO}(2)$, the stabiliser of a point. The SO(2) factor is a circle; its Lie algebra is one-dimensional; call its generator J. On the Hardy space $H^2(D^5_{IV})$ — the physical space of Lecture 3 — J acts with a discrete spectrum, and the eigenvalue of J on a state is that state's conformal weight E. This is the linear conformal

Hamiltonian: the $U(1)$ energy of the representation.

Two things about it are derived and were not obvious to us.

It is J, not the Casimir. An earlier draft took the quadratic Casimir $C_2(K)$ as the time generator. That cannot be right, and the reason is structural: a Casimir is *central* — constant on each irreducible representation — so it labels representations rather than distinguishing states within one; it cannot generate a flow that moves anything. J is linear in the weight and does. The geometry hands us exactly one such operator on the centre of K, so the generator is forced.

Its spectrum is half-integer, with ground weight $E_0 = 3/2$. The building blocks of the representation are the two singletons — the scalar (Rac) at $E = 3/2$ and the spinor (Di) at $E = 2$ — and these weights are read from the literature (Fernando–Günaydin), not chosen. Every weight in the space is $E = \frac{3}{2}\,\#\mathrm{Rac} + 2\,\#\mathrm{Di} + n$ for an integer n. Hold on to the 3/2; it does one job below.

Time is the flow, and the arrow is positivity

The commitment dynamics of Lecture 3 — the contractive face — is the one-parameter semigroup

$$\rho_{\mathrm{commit}}(\tau) \;=\; \exp(-\tau J/\hbar).$$

Time is τ. Not an external axis against which the semigroup is plotted; the semigroup's own parameter, read as "how far the commitment has run." *When* is a measure of accumulated commitment. There is no clock behind this clock.

The arrow is that J is bounded below. $\operatorname{spec} J \geq E_0 = 3/2 > 0$, so $e^{-\tau J}$ is a contraction, defined for $\tau \geq 0$ and not for $\tau < 0$; the flow has no inverse. That is the arrow of time — not an added postulate, but the statement that the energy operator has a ground state and no ceiling. Energy generates, time counts.

We are careful about what kind of arrow this is. It is the *geometric* arrow — the absolute irreversibility of the contractive face. The *thermodynamic* irreversibility of unwriting an amplified record — its cost — is a separate theorem in the companion paper, and we do not merge the two. And it is dynamical, not geometric in the other sense: the shape has no arrow; the dynamics on it does. A CPT-mirror of the substrate — a copy running the other way — is not a second solution that happens not to be realised; it is not a solution at all (the positive-time ontology note). So the program's ontology is pure positive time, and the arrow is not a boundary condition somebody chose.

Charge is not time

Could electric charge be a reading of the *same* circle — charge as where a state sits, time as the circle turning? For a while we thought so. It is excluded, decisively, by degeneracy: the muon and the neutrino are both Rac $\otimes$ Di composites at the same conformal weight $\Delta = 7/2$, and carry $Q = -1$ and $Q = 0$. So Q is not a function of J at all. Electric charge lives in the Cartan of SO(5) — internal — and the SO(2) centre is time's, and only time's (K1687; this was the fifth row that one mislabel had reached). We keep the two words strictly apart: where this lecture says "weight" it means the eigenvalue of J; where it says "charge" it means Q on SO(5).

The two faces

Because J is self-adjoint and bounded below, $\exp(-zJ)$ is a holomorphic semigroup on the right half-plane $\mathrm{Re}\, z \geq 0$ (Hille–Yosida; a standard theorem, cited and not claimed). Its two boundary faces are

- the real-time face $e^{-\tau J}$, $z = \tau$: the irreversible tick, a half-line;
- the imaginary-time face $e^{i\theta J}$, $z = i\theta$: the reversible circle — the unitary evolution of Lecture 3, rule 3.

Euclidean and Lorentzian time are the two edges of one complex parameter, tied by $\tau \leftrightarrow i\theta$. What is specific to this program is not that time has two faces — every bounded-below generator has them — but that the generator is *forced by the geometry*.

Physical time is not periodic. Say this plainly, because the word "circle" invites the misreading. The physical flow is the real face: a one-way half-line with no recurrence. The circle is the Wick face, and its role is a *selection rule* on which states are admissible (next section). We never claim time comes back around.

The circle closes twice — and that is spin-statistics, not a prediction

The imaginary circle's minimal period is fixed by the spectrum of J: within a superselection sector, $T = 2\pi/\gcd\{\text{weights}\}$, and $T = 4\pi$ exactly when the gcd is $\frac{1}{2}$. The Rac's 3/2 supplies the half-integer, so the circle is represented faithfully only on its degree-2 cover: $e^{2\pi i J} = -1$ on half-odd-weight states.

Here is the honest content of that fact. Since $2E \equiv \#\mathrm{Rac} \pmod 2$, one turn acts as $e^{2\pi i J} = (-1)^{\#\mathrm{Rac}}$; on physical particles — two-singleton composites — this is $(-1)^F$, fermion parity, in every dimension. Fermions ride the cover (720°), bosons do not. So the degree-2 cover is the geometric restatement of spin-statistics — a consistency the theory reproduces, not a novel prediction. An earlier draft said it was "forced because $n_C = 5$ is odd"; that was wrong (the odd dimension decides only *which constituent* carries the half-odd weight — here the scalar; in even dimension the spinor — and constituents are not particles). We keep the retraction in print. And the 4π is not observable by interferometry: an interferometer measures phase *differences*, period 2π; the absolute weight that distinguishes 2π from 4π is exactly what it cannot see. Its content is a selection rule, and we say so rather than promise a fringe.

Two ticks, and which is which

Two quantities in this program are called "the tick," and one of the lessons of September was to stop letting one word carry two objects.

The elementary tick of the paper is $\hbar/E$ for the relevant energy: the smallest step of the commitment flow. Its *value* is identified, not derived (the paper's own Section 8 says so), and we do not lean on it.

The Koons tick (T1136) is the program's clock *unit* for writes:

$$\tau_0 \;=\; N_{\max}\,\frac{\hbar}{m_e c^2} \;=\; \frac{a_0}{c} \;\approx\; 0.1765 \text{ attoseconds,}$$

137 reduced Compton times of the electron — equivalently the Bohr radius over the speed of light. It is a *definition*, and its convention matters: a push is 137 radians of the central circle, about 21.8 writes, not 137 turns. One of us wrote "137 turns = 1.1 as" in September; it was a convention slip, caught within the day, and the standing rule that came out of it is to quote the invariant, not the coordinate. The row's own text also once wrote $\nu_C = m_e c^2/\hbar$ for what is an angular frequency; it now reads ω_C.

Because α enters T1136 as a ratio fixed *through the success of a write*, the program's α is constant by definition, and no laboratory drift bound reaches it. We withdrew a claim to the contrary the day it was made (Lecture 8).

What survives a reset

If time is commitment and the universe has cycles, what does the clock carry across a reset? Casey asked this in September and the answer is a floor (K1880–K1884): there is no lawful initialisation vector. A reset that is a law of the substrate is SO(5)-equivariant, and the equivariant survivors of the record space are the constants; the direction average on every saturated state is zero. What survives is a *distribution on the winding count* — four to twelve bits — and nothing lawful consults it. Reentry fails (the environment is legible, not remembered). We had hoped for a richer starting point and did not find one; the floor has three witnesses and stands.

Tier line

- Derived: time as the flow parameter; J as the forced generator; the arrow as positivity; the two Wick faces; charge internal.
- Derived as a consistency, not a prediction: the degree-2 cover is fermion parity.
- Identified: the value of the elementary tick.
- Definition: the Koons tick $\tau_0 = a_0/c$.
- Floored: the nucleation survivor.

- Not claimed: a preferred frame (the corpus derives frame-independence from the group's transitivity on the domain, and the honest form is "the program cannot derive a preferred frame," not "none exists"); periodic physical time; an observable 4π; α drift.

What would make this lecture wrong

A physical process that reverses a committed record at zero cost — an inverse to the contractive face — would falsify the arrow as derived. A boson found to ride the double cover, or a fermion the single one, would break the parity identity, which is protected for free-field singleton weights and *not* protected against anomalous dimensions (the paper says so). Neither is a near-term laboratory test, and we do not dress them as one. The honest exposure is the identified tick value and the open question of what, if anything, the clock's rate has to do with α.

Where to look

"Time, Derived" v1.0 (2026-08-17), Sections 1–8 and its Section 12 — the list of what it does not claim is the paper's strength; K1670 for the pass; Fernando–Günaydin for the weights; T1136 with its 09-07 annotations; K1687; K1880–K1884 for the reset floor. One alias to carry: the paper's abstract writes "$H^2(D^5_{IV})$, the Bergman space" — read *Hardy* (the W1 resolution came five days after the paper).